



Geometry Quest

The mission where shapes become claims you can check.

THE BIG QUESTION

Two shapes can have the same perimeter and different areas. How?

HANDS-ON PROJECT

Map a Planet

Invent a world on the coordinate plane, then give sailing directions using nothing but coordinates.

FLUENCY GAME

Angle Guess

Estimate an angle, then measure it. Closest guess scores the round.



AXIOM • MISSION COMMANDER, NINE TOURS

“Brock — navigation is geometry with consequences. Across first, then up. Say it until it's boring.”

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. You'll need a protractor for two of these — estimate first, then measure.

STRAND A ANGLE FACTS

A1 How many degrees in a right angle?
In a full turn?

.....

A2 Two angles on a straight line. One is
130°. The other?

.....

STRAND B MEASURING

B1 Draw an angle of about 45°. Then
measure it.



B2 Is an angle of 100° bigger or smaller
than a right angle?

.....

STRAND C PERIMETER

C1 Perimeter of a 5 by 3 rectangle?

.....

C2 Area of the same rectangle?

.....

STRAND D COORDINATES

D1 Which number in (3, 5) tells you how
far across?

.....

D2 How far is it from (2, 3) to (7, 3)?

.....

STRAND E SORTING SHAPES

E1 Is a square also a rectangle? Explain.

.....

E2 How many sides has a hexagon? A
pentagon?

.....

STRAND F SYMMETRY

F1 How many lines of symmetry has a
square?

.....

F2 And a rectangle that is not a square?
Why fewer?

.....

Skip Chart

Score the pre-assessment, then cross days off the five-week calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A • Angle facts	A1 A2	Week 1 • lessons 1.1 and 1.3	--- / 2	Day 1.1. Warm-Up tier drops to 3 items all unit.
B • Measuring	B1 B2	Week 1 • lesson 1.2	--- / 2	Day 1.2 — but keep Angle Guess, it's five minutes and it builds the estimate habit.
C • Perimeter & area	C1 C2	Week 2 • lessons 2.1–2.2	--- / 2	Days 2.1–2.2. Never skip 2.3 — that lesson is the Big Question.
D • Coordinates	D1 D2	Week 3	--- / 2	Days 3.1–3.3. Keep 3.4 and the whole project.
E • Sorting shapes	E1 E2	Week 1 • lesson 1.4	--- / 2	Day 1.4. E1 answered “no” means keep it whatever E2 says.
F • Symmetry	F1 F2	Week 4 • lessons 4.1–4.2	--- / 2	Days 4.1–4.2. Keep 4.3, flips and turns, and the tessellation Friday.

Where the saved time goes

Into Map a Planet. A bigger grid, a second navigator, or a route that has to avoid hazards all beat another worksheet. Write the reallocation here:

.....
.....

Revised Mission 06 plan

WEEK 1 ☐ full ☐ compressed ☐ skip
WEEK 2 ☐ full ☐ compressed ☐ skip
WEEK 3 ☐ full ☐ compressed ☐ skip
WEEK 4 ☐ full ☐ compressed ☐ skip
WEEK 5 ☐ full ☐ compressed ☐ skip

What an Angle Is

Not a corner. An amount of turn, measured in degrees.

✦ Warm-Up

DEGREES, FROM MEMORY

a right angle

a straight line

a full turn

right angles in a full turn

half a right angle

right angles in a straight line

THE FOUR FACTS

right angle = 90°
straight line = 180°
full turn = 360°
triangle = 180°

Almost every angle question this week is one of these four minus something you already know. Reach for the arithmetic before the protractor.

◆ Core

WORK IT OUT — NO PROTRACTOR

1. Two angles on a straight line. One is 130°. The other is _____°
.....
2. Three angles round a point: 90°, 120° and _____°
.....
3. A triangle has angles 60° and 70°. The third is _____°
.....
4. An angle of 200° is how far past a straight line?
.....
5. Two equal angles sit on a straight line. Each one is _____°
.....

★ Challenge

NOT GRADED

C1. Add up all four angles inside a square. Then do the same for any rectangle.
.....

C2. Turn 90° four times. How far have you turned altogether, and where are you facing?
.....

C3. Can a triangle have two right angles? Try to draw one, then explain what goes wrong.
.....

 **AXIOM:** Estimate before you measure. If the protractor disagrees with your estimate by more than ten degrees, one of you has made a mistake — find out which.

Perimeter & Area

Perimeter is the fence. Area is the grass. Knowing one never tells you the other.

Warm-Up READ CAREFULLY — WHICH ONE IS BEING ASKED FOR?

perimeter of 5×3

area of 5×3

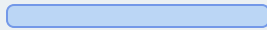
perimeter of square side 6

area of square side 6

perimeter of 10×2

area of 10×2

TWENTY-FOUR METRES OF FENCE, FOUR DIFFERENT FIELDS



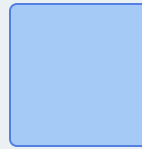
$$1 \times 11 = 11$$



$$2 \times 10 = 20$$



$$4 \times 8 = 32$$



$$6 \times 6 = 36$$

Every field above has exactly 24 metres of fence. The grass runs from 11 to 36 square metres. The square wins — and it always does.

Core

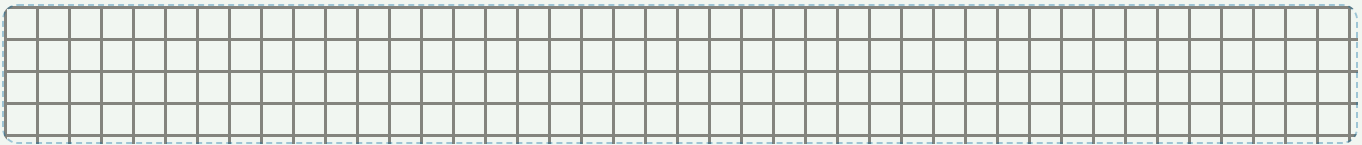
1. Area 48, one side 6. The other side?

2. Perimeter 30, one side 9. The other?

3. Area of a 12 by 7 rectangle

4. Perimeter of the same 12 by 7

5. An L-shape made of a 6 by 4 block joined to a 3 by 2 block. Draw it, cut it into two rectangles, and find the total area.



Challenge

C1. List every rectangle with perimeter 24 and whole-number sides. Which has the biggest area? The smallest?

.....

C2. You have 24 metres of fence *and* a long wall to build against, so you only need three sides. Now what's the biggest field?

.....

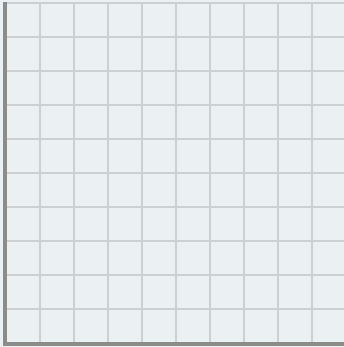
The Coordinate Plane

Across first, then up. (3, 5) is not (5, 3).

Warm-Up WHICH NUMBER IS THE ACROSS ONE?

(3, 5)
_____(7, 2)
_____(0, 4)
_____(6, 6)
_____(9, 1)
_____(5, 0)

PLOT THESE SIX POINTS



A (3, 5) B (5, 3) C (0, 4)

D (7, 7) E (9, 0) F (6, 6)

Label each one. Then answer: which two points would swap places if you read the numbers the wrong way round?

.....

Core

1. Distance from (2, 3) to (7, 3)

2. Distance from (4, 1) to (4, 8)

3. Rectangle (1,1) (6,1) (6,4) (1,4) — perimeter

4. That same rectangle — area

5. Start at (2, 2). Go 4 across and 3 up. Where do you land?

Challenge

C1. Three corners of a square are (0,0), (5,0) and (5,5). Where is the fourth?

.....

C2. Write a three-leg route as coordinates only, then have someone follow it without seeing your map.

.....

Sorting Shapes & Symmetry

Sort by what is true about a shape, not by what it looks like.

✦ Warm-Up

sides on a
hexagon

sides on a
pentagon

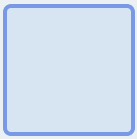
right angles in a
rectangle

equal sides on
an equilateral
triangle

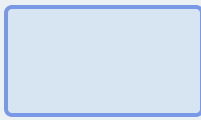
sides on an
octagon

pairs of parallel
sides in a
rectangle

FOLD IT AND SEE • DRAW EVERY LINE OF SYMMETRY



square ___



rectangle ___



triangle ___



circle ___

A fold is a line of symmetry only if the two halves land exactly on top of each other. Try the corner folds as well as the middle ones.

◆ Core

1. Lines of symmetry in a square

2. Lines in a rectangle that isn't square

3. Lines in an equilateral triangle

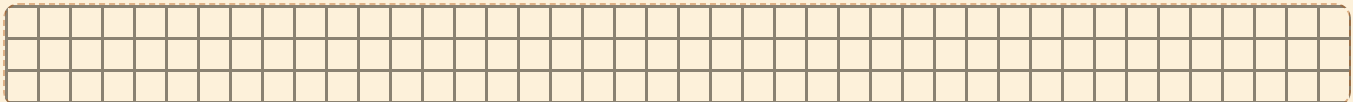
4. Lines in a circle

5. Is a square also a rectangle? Is a rectangle also a square? Answer both, and say why the two answers differ.

★ Challenge

C1. How many lines of symmetry has a regular hexagon? Find the pattern for regular shapes.

C2. Draw a four-sided shape with exactly one line of symmetry. What is it called?



Angle Guess

Estimate first. Measure second. The gap between them is the score.

The game • two players, one protractor

One player draws any angle without measuring it. Both players write down an estimate. Then measure. **Your score is how far off you were** — lowest total after eight rounds wins. Take turns drawing.

ROUND	MY ESTIMATE	MEASURED	OFF BY	ACUTE, RIGHT OR OBTUSE?
1				
2				
3				
4				
5				
6				

★ Puzzles FOUR FACTS, NO PROTRACTOR

On a line next to 47°

.....

Round a point: 100° , 130° , ?

.....

Perimeter 20, biggest area

.....

Area 36, smallest perimeter

.....

(3,7) to (11,7)

.....

Turn 45° eight times

.....

Angles in any triangle

.....

Lines of symmetry, regular hexagon

.....

The real question

Which shapes tile a floor with no gaps at all? Test a square, a regular hexagon and a regular pentagon. Two of them work and one doesn't — explain what goes wrong with the one that fails, using the angles at the corner.

.....

.....

The Rest of the Mission

Same three-tier shape as Week 1, every day. Objectives outlined here; worksheets follow the Week 1 pattern.

Week 2 · Perimeter & Area

THE BIG QUESTION GETS ANSWERED THIS WEEK

2.1 Mon

Perimeter is the fence, area is the grass.

2.2 Tue

Area of rectangles, with units attached.

2.3 Wed

Same fence, different grass — every rectangle with perimeter 24.

2.4 Thu

Compound shapes: cut the L into two rectangles.

Fri

The biggest pen, using a wall as one side.

Week 3 · The Coordinate Plane

MID-UNIT QUIZ FRIDAY • 85% TO ADVANCE

3.1 Mon

Across, then up. Plot (3,5) and (5,3) side by side.

3.2 Tue

Plot a shape; read its sides off the grid.

3.3 Wed

Distance on the grid by subtracting.

3.4 Thu

Sailing directions as a list of coordinates.

Fri

Mid-unit quiz, 8 items, Weeks 1–3.

Week 4 · Symmetry & Transformation

THE PRETTIEST WEEK OF THE MISSION

4.1 Mon

Lines of symmetry, found by folding.

4.2 Tue

Why a square has four and a rectangle two.

4.3 Wed

Flips and turns on squared paper.

4.4 Thu

Five symmetric things he actually owns.

Fri

Tessellation: which shapes tile, and why.

Week 5 · Proof & Project

UNIT TEST FRIDAY • BADGE AWARDED

5.1 Mon

Design the planet: coastline and three landmarks, plotted.

5.2 Tue

Write the route by coordinates only.

5.3 Wed

Sail someone else's map and compare.

Thu

Error journal sweep.

Fri

Mission 06 test: 12 items + one explanation.

MATERIALS, WHOLE UNIT

A real protractor • ruler • squared paper • scissors and tracing paper for the folding • coloured pencils • the error journal

WEEKLY CHECK

Five items every Friday, mixed tiers. Below 4/5 means Monday is a reteach, not new material.

THE MOST COMMON STALL

Reading the protractor off the wrong scale, so 40° comes out as 140°. The estimate catches it every time — which is why the estimate comes first.

Mid-Unit Quiz

8 items • Weeks 1–3 • 7 of 8 keeps you flying

____ / 8

1. Two angles on a straight line. One is 47° . The other?

.....

2. A triangle has angles 60° and 70° . The third?

.....

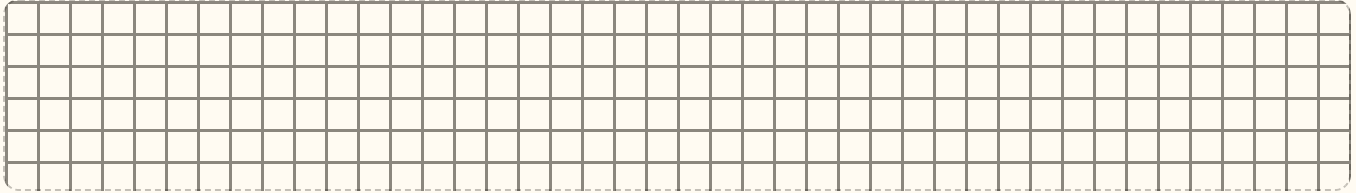
3. Perimeter and area of a 12 by 7 rectangle.

.....

4. Area 48, one side 6. How long is the other?

.....

5. Two rectangles both have perimeter 24. Draw one with area 20 and one with area 36.



6. How far is it from (2, 3) to (7, 3)?

.....

7. Start at (2, 2), go 4 across and 3 up. Where do you land?

.....

CHALLENGE ITEM

8. A rectangle has perimeter 24 and sides 8 and $\square$. What is $\square$, and what is the area?

.....

Mission 06 Test

12 items + one explanation · part 1 of 2

1. Degrees in a full turn

.....

2. On a line beside 115°

.....

3. Triangle: 45° , 85° , ?

.....

4. Draw an angle of about 130° , then measure it. Write your estimate and the measurement.**5.** Perimeter and area of a 9 by 4 rectangle.

.....

6. Area 56, one side 7. The other side?

.....

7. An L-shape: a 6 by 4 block plus a 3 by 2 block. Total area?

.....

8. Distance from (1, 2) to (1, 9)?

.....

9. Three corners of a square are (0,0), (5,0) and (5,5). Give the fourth, then the square's perimeter.

.....

Reasoning & the Big Question

10. Lines of symmetry in a square, and in a rectangle

.....

11. Is a square also a rectangle? Why?

.....

12. Angles round a point: 100° , 130° , ?

.....

CHALLENGE ITEM

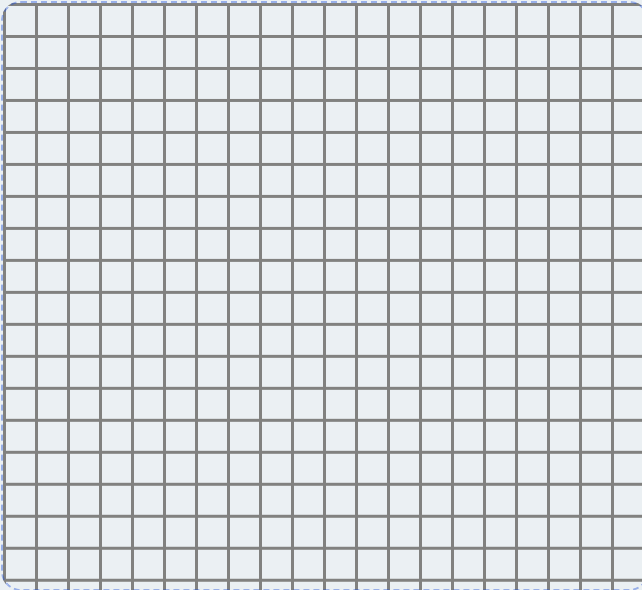
You have 20 metres of fence. What is the biggest rectangular field you can build with whole-number sides — and what if a wall gives you one side free?

.....

EXPLAIN YOUR THINKING • WORTH 4 POINTS

Two shapes can have the same perimeter and different areas.
How?

Draw two rectangles with the same perimeter and clearly different areas. Label both. Then say which shape gives the biggest area for a fixed perimeter, and why. Spelling doesn't count. Reasoning does.



.....

.....

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.....

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.....

TROPHY BAND • CIRCLE ONE

85–100%

Mission Commander

Compass Badge earned. Mission 07 opens — and Challenge becomes your Core.

70–84%

Flight Engineer

Two targeted days on the flagged items, then retake those items only.

Below 70%

Ground Crew

Back to squared paper and physical folding before any more symbol work.

Key A • Pre-Assessment, Lessons 1.1–1.2

Pre-Assessment

- A1 90° and 360°
 A2 50°
 B1 measurement within 10° of the drawing
 B2 **bigger** – obtuse
 C1 16 C2 15
 D1 the **first** number, 3
 D2 5
 E1 **yes** – four right angles and opposite sides equal
 E2 6 and 5
 F1 4
 F2 2 – the corner folds don't match

Scoring note. E1 answered “no” is the classic sorting error — he is sorting by appearance rather than by properties. Run day 1.4 whatever else he scored.

Lesson 1.1 • What an Angle Is

- Warm-Up** 90 • 180 • 360 • 4 • 45 • 2
Core 1. 50° 2. 150° 3. 50°
 4. 20° 5. 90° each

Lesson 1.1 Challenge. C1 • 360° for a square and for any rectangle — four right angles. C2 • 360° altogether, and you're facing exactly where you started. C3 • no. Two right angles already use up all 180° , leaving nothing for the third angle, so the other two sides would never meet.

Lesson 1.2 • Perimeter & Area

Warm-Up 16 • 15 • 24 • 36 • 24 • 20

Core

1. 8 2. 6
 3. area 84 4. perimeter 38
 5. $24 + 6 = 30$ square units

Challenge. C1 • 1×11 (11), 2×10 (20), 3×9 (27), 4×8 (32), 5×7 (35), 6×6 (36). Biggest is the square at 36, smallest is 1×11 at 11. C2 • with a wall on one side you're fencing only three sides, so $24 = 2$ depths + 1 length; the best is 6 deep by 12 long, giving **72** square metres — double the closed-field answer. That result surprises everyone, and it's worth sitting with.

Watch for: adding all four sides to find area, or multiplying to find perimeter. Ask “fence or grass?” before he writes anything.

Key B • Lessons 1.3–1.4, Enrichment, Quiz

Lesson 1.3 • The Coordinate Plane

Warm-Up (across numbers) 3 • 7 • 0 • 6 • 9 • 5

Plotting A and B swap; the rest are unaffected

Core

1. 5 2. 7

3. perimeter 16 4. area 15

5. (6, 5)

Challenge. C1 • (0, 5). C2 • mark the route correct if the follower's finished path matches the author's map exactly. When they disagree, the interesting question is always which of the two wrote it down wrong — make him find out rather than telling him.

Lesson 1.4 • Sorting & Symmetry

Warm-Up 6 • 5 • 4 • 3 • 8 • 2

Shapes box square 4 • rectangle 2 •

equilateral triangle 3 • circle infinitely many

Core 1. 4 2. 2 3. 3

4. infinitely many

5. a square is always a rectangle; a rectangle is only sometimes a square

Challenge. C1 • six, and a regular shape with n sides has n lines of symmetry. C2 • a kite has exactly one; an isosceles trapezium also works. Accept either with a correct drawing.

Friday • Angle Guess

Puzzles

on a line beside $47^\circ \rightarrow 133^\circ$

round a point $\rightarrow 130^\circ$

perimeter 20, biggest area $\rightarrow 25$ (5×5)

area 36, smallest perimeter $\rightarrow 24$ (6×6)

(3,7) to (11,7) $\rightarrow 8$

45° eight times $\rightarrow 360^\circ$

triangle angles $\rightarrow 180^\circ$

regular hexagon $\rightarrow 6$

The real question. Squares tile (four 90° corners meet to make 360°) and regular hexagons tile (three 120° corners make 360°). Regular pentagons do not: each corner is 108° , and three of them make 324° — leaving a 36° gap that no number of pentagons can fill. Accept any answer that gets to “the corner angles have to add to exactly 360° .”

Mid-Unit Quiz • out of 8

1. 133° 2. 50°

3. perimeter 38, area 84

4. 8

5. 2×10 and 6×6 (accept any correct pair)

6. 5 7. (6, 5)

8. $\square = 4$, area 32

7/8 or better $\rightarrow$ continue. A miss on 3 is the perimeter/area mix-up and needs one targeted day. A miss on 7 means the coordinate order hasn't stuck — fix it before Week 5, because the whole project depends on it.

Key C • Mission 06 Test & Scoring

Items 1–12 • one point each

1. 360°
2. 65°
3. 50°
4. drawing within 10° of 130° , plus a measurement
5. perimeter **26**, area **36**
6. **8**
7. $24 + 6 = 30$
8. **7**
9. **(0, 5)**, perimeter **20**
10. **4** and **2**
11. **yes** – it has four right angles and opposite sides equal, which is the whole definition
12. 130°

Challenge item. Closed field: $5 \times 5 = 25$. With a free wall: 5 deep by 10 long = **50**.

Explanation • 4 points

Score the reasoning, not the writing. Award one point each:

- ☐ 1 • Two rectangles drawn with the same perimeter
- ☐ 2 • Both areas correctly computed and clearly different
- ☐ 3 • Identifies the square as the biggest-area shape
- ☐ 4 • Explains that the fence measures the edge, not the inside

Model answer, kid-sized. “A 2 by 10 and a 6 by 6 both need 24 metres of fence, but one holds 20 square metres and the other holds 36. The fence only measures the edge. Long thin shapes waste fence going out and back, so the square holds the most.”

Total 16 points. 14+ Mission Commander • 11–13 Flight Engineer • 10 or fewer Ground Crew. Item 4 is scored on the estimate-then-measure habit, not on precision; anything within 10° earns the point.

Error Journal • photocopy this page

One row per mistake, for the whole mission. The “why” line must name a step. “I was careless” doesn't count.

DATE	PROBLEM	I GOT / ANSWER	WHY IT HAPPENED – ONE SENTENCE